

Integrated Space Logistics Mission Planning and Spacecraft Design with Mixed-Integer Nonlinear Programming

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This paper develops a campaign-level space logistics optimization framework that simultaneously considers mission planning and spacecraft design using mixed-integer nonlinear programming. In the mission planning part of the framework, deployment and utilization of in-orbit infrastructures, such as in-orbit propellant depots or in situ resource utilization plants, are also taken into account. Two methods are proposed: First, the mixed-integer nonlinear programming problem is converted into a mixed-integer linear programming problem after approximating the nonlinear model with a piecewise linear function and linearizing quadratic terms. In addition, another optimization framework is provided, based on simulated annealing, which separates the spacecraft model from mission planning formulation. An example mission scenario based on multiple Apollo missions is considered, and the results show a significant improvement in the initial mass in low Earth orbit by campaign-level design as compared with the traditional mission-level design. It is also shown that the mixed-integer linear programming-based method gives better-quality solutions than the simulated annealing-based method, although the simulated annealing method is more flexible for extension to a higher-fidelity spacecraft model.

Nomenclature

A	=	set of directed arcs
b	=	spacecraft flow vector (binary)
C	=	spacecraft payload capacity
f	=	spacecraft fuel type (integer)
G	=	network graph
g_0	=	gravitational acceleration on Earth (equal to 9.8 m/s^2)
H	=	concurrence matrix
I_{sp}	=	specific impulse
i	=	node index ($\in \mathcal{N}$)
$\mathcal{J}$	=	objective function
j	=	node index ($\in \mathcal{N}$)
M	=	spacecraft propellant capacity
M_O	=	oxygen in situ resource utilization plant mass
M_{UB}	=	upper bound of propellant capacity
N	=	set of nodes
N_O	=	oxygen in situ resource utilization plant production
N_M	=	oxygen in situ resource utilization plant productivity
O	=	continuous variable denotes payload or propellant capacity
P	=	polyhedron formed by epigraph of piecewise function
$\mathcal{P}$	=	union of polyhedra
Q	=	transformation matrix
R	=	recession cone
s	=	spacecraft structure mass
t	=	time index (integer)
t_b	=	spacecraft impulsive burn time
U	=	branching scheme
V	=	set of spacecraft
$\mathcal{V}$	=	set of breakpoints
v	=	spacecraft-type index
W	=	set of time windows

x	=	payload flow vector
y	=	spacecraft flow variable (integer)
α	=	structural fraction
ϕ	=	propellant mass fraction

I. Introduction

AS SPACE exploration projects become increasingly complex, the campaign-level logistics perspective becomes more important for human space mission planning. In many past human space programs such as Apollo, every mission was logistically independent using a so-called carry-along strategy. This was efficient for short-term missions to a relatively close destination. However, for a longer-duration multimission space exploration campaign, reusable in-orbit infrastructures can be more attractive. For example, it can be beneficial to have an in-orbit propellant depot and/or a lunar in situ resource utilization (ISRU) plant to refuel human missions to Mars or near-Earth objects (NEOs) in return for their emplacement cost and the required maintenance. There are pros and cons for operation of those infrastructures. Therefore, it is important to perform an integrated campaign-level analysis considering both mission planning and infrastructure design to evaluate whether it pays off to deploy those infrastructures.

In addition, we need a paradigm shift in vehicle design as well. The conventional space programs such as Apollo delivered relatively similar payloads to the destination of each mission throughout the campaign, and therefore a few identical single-use vehicles were sufficient for their logistics. For example, each Apollo vehicle carried a similar amount of similar types of commodities (e.g., consumables, scientific payloads) in all missions. However, in a longer-duration multimission space exploration campaign with infrastructure deployment and utilization, we need to deliver different types and amounts of commodities to the destination throughout the campaign. For instance, in the beginning of the campaign, the spacecraft may need to carry ISRU plants, whereas later on in the campaign, it may need to carry the ISRU-generated propellants and its tank structure. Thus, for such campaign-level mission design, it is critical to study and analyze the optimal fleet of multiuse vehicles, where each vehicle is flexible in its assigned function.

Many past studies have addressed space logistics mission planning, but few of them have considered the in-orbit infrastructure design and spacecraft model concurrently at a reasonable level of fidelity. One of the most recent studies that analyzed space mission optimization was the generalized multicommodity network flow (GMCNF) formulation developed by Ishimatsu et al. [1]. Their research modeled the space logistics planning problem as a linear

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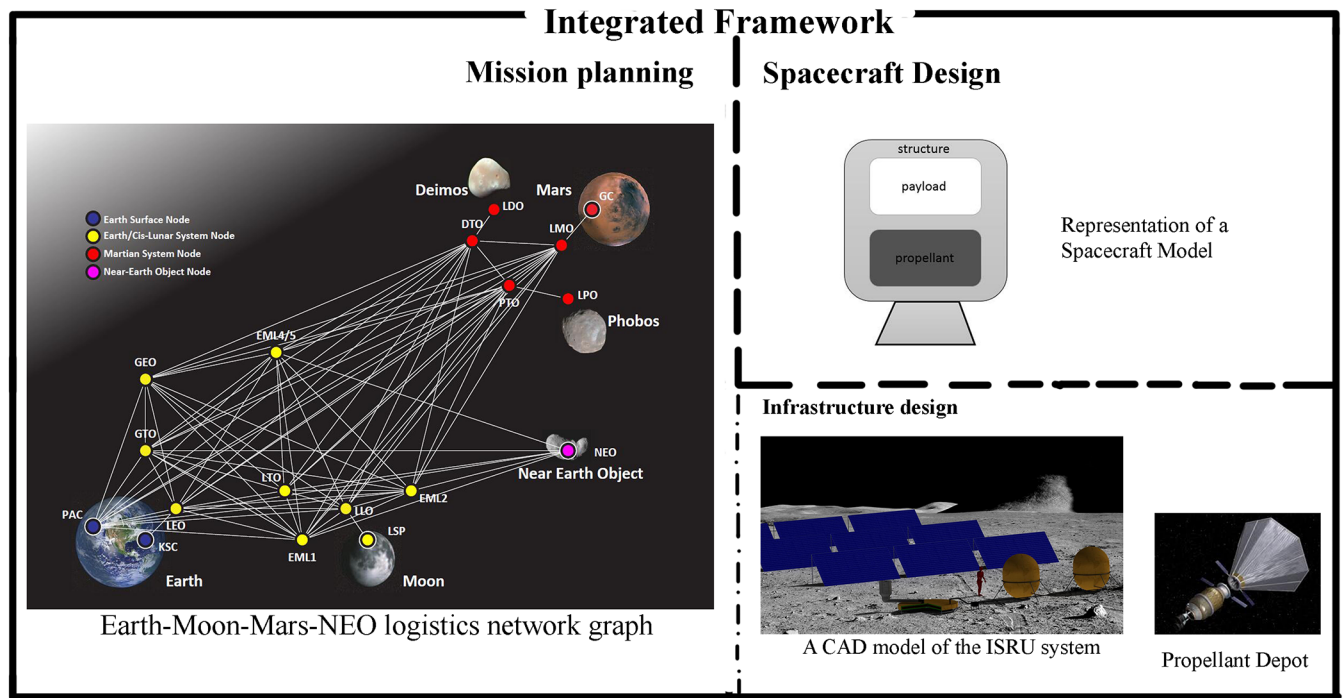


Fig. 1 Integrated optimization framework (adapted from [7,15–17]).

programming problem using a graph-theoretic approach, but this method did not include the emplacement cost for the in-orbit infrastructures properly. Ho et al. improved the GMCNF by proposing a dynamic space logistics optimization formulation that removed “time paradoxes” existing in the static network [2], referring to the propellant produced by ISRU that could be used before the ISRU plants were deployed. In addition to this, Ho et al. considered the emplacement cost of the in-orbit infrastructure. However, the infrastructure and spacecraft model used in the analysis was a low-fidelity linear model, which could result in an unrealistic solution (e.g., an unrealistically small vehicle size). Other space logistics optimization models such as the interplanetary logistics model [3], the exploration architecture model for in-space and Earth-to-orbit tool [4], and the space system architectures model based on graph theory [5] have been proposed for space logistics mission planning, but these studies did not consider the possibilities of using propellant depots and ISRU plants, which can be critical for long-term space exploration.

On the other hand, various studies have focused on the detailed design of in-orbit infrastructures, but few of them have also addressed global optimization of space mission planning concurrently. For example, Oeftering [6] studied detailed cislunar infrastructure design and Schreiner [7] performed a detailed system-level ISRU plant sizing analysis for the molten regolith electrolysis process. Also, there were numerous architectures proposed about on-orbit propellant depots from both technological perspectives [8–10] and economical perspectives [11–13]. However, those studies only looked at the design of the depot with its local transportation system rather than providing a design method considering global optimization for the entire space logistics system.

As shown previously, most of the past space logistics literature has not considered space mission planning, in-orbit infrastructure, and vehicle design concurrently at a reasonable fidelity. In response to this background, this paper proposes a system-level optimization formulation, considering mission planning, in-orbit infrastructure design, and spacecraft design simultaneously, aiming to evaluate the cost and benefit of reusable in-orbit infrastructures and generalized designed spacecraft. The model is a mixed-integer nonlinear programming formulation based on a time-expanded GMCNF model. This model significantly advances Ho et al.’s previous model [2] and takes into account higher-fidelity in-orbit infrastructure design models, including propellant depots and ISRU. We also add

new constraints for vehicles and crew members. The integrated optimization framework is shown in Fig. 1. This optimization framework consists mainly of two parts: mission planning including infrastructure design, and spacecraft design. Note that propellant depot design is not considered with a specific model in this paper; instead, the spacecraft model is used for depot design, assuming a spacecraft staying in a certain orbit can serve as a temporary propellant depot. A similar concept to evolve the spacecraft to a propellant depot was proposed by the United Launch Alliance [14].

Our work will provide an important step to close the gap in the past literature and propose an integrated optimization formulation, considering both mission planning (including in-orbit infrastructure design) and spacecraft design using a mixed-integer programming formulation. The developed formulation can be particularly useful for evaluating the cost and benefit of cislunar infrastructure and for developing concrete space mission schedules for future long-term space exploration.

II. Methodology

The optimization formulation, aiming to minimize the total cost of the whole multi-mission campaign, is developed based on Ho et al.’s time-expanded generalized multicommodity network flow model [2]. To improve the fidelity of the infrastructure modeling, we take into account a higher-fidelity ISRU model based on Schreiner system-level ISRU plant sizing analysis [7], a high-fidelity spacecraft model, and new constraints for crew members.

This integrated optimization problem turns out to be a complex mixed-integer nonlinear programming (MINP) problem and cannot be solved using the conventional formulations proposed by Ishimatsu et al. [1] and Ho et al. [2] due to their linear programming natures. The nonconvex nonlinear in-orbit infrastructure design model, which specifically denotes ISRU in this paper; the complex nonlinear spacecraft design model; and the quadratic terms in both the objective function and constraints make it very challenging to solve the problem and find the global optimum directly [3].

To solve the optimization problem efficiently, we developed two different methods.

The first method to solve this problem converts the whole problem into a MILP problem. The nonlinear ISRU and spacecraft models are approximated by piecewise linear functions, to then be converted into

a binary mixed-integer programming formulation. The remaining quadratic terms in the mission planning formulation are linearized. This MILP formulation can always find the approximated global optimum. The detail of this MILP optimization method is introduced in Sec. II.B.2.

Another developed method solves mission planning and the spacecraft model separately using simulated annealing (SA). Because the spacecraft model is separated from mission planning, no quadratic term exists in the formulation. Only the ISRU model needs to be approximated by a piecewise linear function. This SA-based method cannot guarantee its global optimality, but it can potentially provide a solution to the problem with a more realistic and complex spacecraft model. The detail of this SA-based optimization framework is shown in Sec. II.B.3.

A. Modeling

This section introduces the space logistics, spacecraft, and ISRU modeling methods. The space logistics model is developed based on Ho et al.'s time-expanded GMCNF [2] while using higher-fidelity spacecraft and ISRU models. The spacecraft model used in this paper is developed based on Taylor's integrated transportation system design framework [15]. This is a data-based model where all the data come from preexisting spacecraft. The ISRU plant considered in this paper is an oxygen ISRU developed by Schreiner [7] based on their system-level ISRU plant sizing analysis for the molten regolith electrolysis process.

1. Space Logistics Modeling

The space logistics problem can be solved as a multicommodity network flow problem. In the network, nodes correspond to planets, celestial objects, or orbits; and arcs correspond to trajectories. Spacecraft, crew members, scientific instruments, and other kinds of payloads are considered as commodities flowing along arcs. The left-hand side of Fig. 1 shows a network graph of the Earth-moon-Mars-NEO space logistics model [16–18]. A generalized multicommodity network flow formulation has been developed with this network graph for space logistics optimization [1,2].

The space logistics model in this paper is developed based on a time-expanded GMCNF formulation. Solutions of the time-expanded GMCNF provide schedules that determine when to transport which commodity from one node to another. In past research, everything that flowed over arcs was considered to be one kind of commodity, including crew members, consumables, ISRU plants, spacecraft, etc. [1,2]. This paper separates the spacecraft from other commodities, and its model is considered as a separate module so that its nonlinear mixed-integer feature can be included efficiently.

The following shows the mathematic formulation of time-expanded GMCNF. Consider a time-expanded network graph G that is made up of a set of nodes $\mathcal{N}$ and a set of direct arcs A , including both transportation arcs that connect different nodes and holdover arcs that connect the node and itself. Each arc has an index (v, i, j) , meaning that spacecraft v flies from node i to node j . We assume that the spacecraft set is V . Commodities flow over arcs and are split into outflow $\mathbf{x}_{vijt}^+$ and inflow $\mathbf{x}_{vijt}^-$, where t is the time step. There are also cost coefficients c_{vijt}^+ assigned to outflow. Note that, if we assume there are p types of commodities delivered in spacecraft, $\mathbf{x}_{vijt}^+$ and c_{vijt}^+ are all $p \times 1$ vectors, where each component shows the flows and costs of the corresponding commodity. Therefore, all flows $\mathbf{x}_{vijt}^\pm$ are nonnegative. Most of the commodities in $\mathbf{x}_{vijt}^\pm$ are continuous variables, except crew members, which form a discrete (i.e., integer) variable. Moreover, $\mathbf{x}_{vijt}^\pm$ represents those commodities delivered by spacecraft. The spacecraft itself is also one kind of commodity denoted by $y_{vijt}^\pm$, which is a discrete (i.e., integer) variable.

Each node i has a demand vector $\mathbf{d}_{it}$, which is the demand or supply of each commodity at time step t . In the demand vector, demand is shown by a negative value, whereas supply is shown by a positive value. Figure 2 shows an example of a full-time expanded network. For instance, the commodity in node k can take one of three paths:

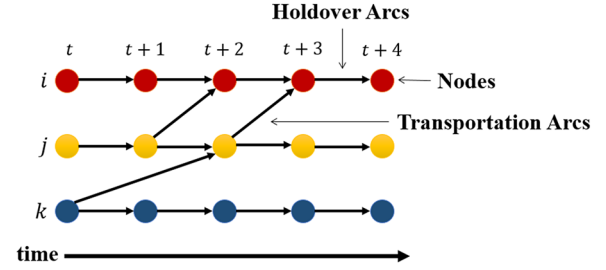


Fig. 2 Full time-expanded network [16].

- 1) Stay at node k over the holdover arc until $t + 4$.
- 2) Be transported to node j at time t , and stay at node j until $t + 4$.
- 3) Be transported to node j at time t , and then to node i at time $t + 2$, and stay at node i until $t + 4$.

The optimizer will choose the optimal path of these three.

For each arc from node i to node j , a positive transit time Δt_{ij} is defined. Besides transit time, another important parameter of time-expanded GMCNF is the possible departure time $W_{ij} \subseteq [0, T)$, which corresponds to the time windows of arc i to j . T denotes the maximum time horizon of the campaign-level sequence of space missions. The concept of the time window defines whether it allows commodity flows at a specified time step.

According to the aforementioned notations, the time-expanded GMCNF can be expressed by the following formulation:

Minimize

$$\mathcal{J} = \sum_{t \in \{0, \dots, T-1\}} \sum_{(v, i, j) \in A} \left(c_{vijt}^+ \mathbf{x}_{vijt}^+ + c_{vijt}' s_v y_{vijt}^+ \right) \quad (1)$$

subject to

$$\sum_{(v, j) : (v, i, j) \in A} \mathbf{x}_{vijt}^+ - \sum_{(v, i) : (v, i, j) \in A} \mathbf{x}_{vijt}^- \leq \mathbf{d}_{it} \quad \forall t \in \{0, \dots, T-1\} \quad \forall i \in N \quad (2)$$

$$\sum_{j : (v, i, j) \in A} y_{vijt}^+ - \sum_{j : (v, j, i) \in A} y_{vijt}^- \leq d'_{ivt} \quad \forall t \in \{0, \dots, T-1\} \quad \forall i \in N \quad \forall v \in V \quad (3)$$

$$Q_{vij} \begin{bmatrix} \mathbf{x}_{vijt}^+ \\ s_v y_{vijt}^+ \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{vijt}^- \\ s_v y_{vijt}^- \end{bmatrix} \quad \forall t \in \{0, \dots, T-1\} \quad \forall (v, i, j) \in A \quad (4)$$

$$H_{vij} \mathbf{x}_{vijt}^+ \leq e_v y_{vijt}^+ \quad \forall t \in \{0, \dots, T-1\} \quad \forall (v, i, j) \in A \quad (5)$$

$$\begin{cases} \mathbf{x}_{vijt}^\pm \geq 0_{p \times 1} & \text{if } t \in W_{ij} \\ \mathbf{x}_{vijt}^\pm = 0_{p \times 1} & \text{otherwise} \end{cases} \quad \forall t \in \{0, \dots, T-1\} \quad \forall (v, i, j) \in A \quad (6)$$

$$s_v = \mathcal{F}(e_v, f_v) \quad \forall v \in V \quad (7)$$

$$\mathbf{x}_{vijt}^\pm = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}_{vijt}^\pm \quad \text{where } x_1 \in \mathbb{Z}_+, x_2, \dots, x_p \in \mathbb{R}_+$$

$$\forall t \in \{0, \dots, T-1\} \quad \forall (v, i, j) \in A$$

$$y_{vijt}^\pm \in \mathbb{Z}_+ \quad \forall t \in \{0, \dots, T-1\} \quad \forall (v, i, j) \in A$$

$$s_v \in \mathbb{R}_+, \quad e_v \in \mathbb{R}_+^l, \quad f_v \in \mathbb{Z}_+ \quad \forall v \in V$$

Table 1 Definitions of indices, variables, and parameters

Name	Definition (dimension)
<i>Index</i>	
v	Spacecraft index
i, j	Node
t	Time step
p	Commodity index
l	Concurrency constraint index
<i>Variables</i>	
$x_{vijt}^{\pm}$	Commodity outflows/inflows. Commodities in $x_{vijt}^{\pm}$ are considered as integer or continuous variables based on the commodity type. For example, the number of a crew member is an integer variable, which is x_1 , whereas other commodities (such as propellant, payload, and human consumables) are considered as continuous variables, which are $x_2, \dots, x_p$. ($p \times 1$)
$y_{vijt}^{\pm}$	Number of spacecraft v flying along arc i to j . Integer variable. (scalar)
s_v	Structure mass of spacecraft v . Continuous variable. (scalar)
e_v	Spacecraft design parameters, including payload capacity and propellant capacity. Continuous variables. ($l \times 1$)
f_v	Spacecraft fuel type. Integer variable. (scalar)
<i>Parameters</i>	
c_{vijt}^+	Commodity cost coefficient. ($p \times 1$)
c_{vijt}'	Spacecraft cost coefficient. (scalar)
d_{it}	Demands or supplies of different commodities at each node. ($p \times 1$)
d_{it}'	Demand or supply of spacecraft v at each node. (scalar)
Q_{vij}	Commodity transformation matrix. ($(p+1) \times (p+1)$)
H_{vij}	Concurrency constraint matrix. ($l \times p$)
W_{ij}	Time window vector. ($1 \times n$, where n is the number of time windows. It is dependent on space missions.)

Table 1 lists the definitions of the indices, variables, and parameters. The detailed descriptions of the objective function and constraints are as follows:

Equation (1) is the objective function. It gives the value of the initial mass in low Earth orbit (IMLEO), which is the cost metric for space logistics in this paper.

Equations (2) and (3) are mass balance constraints that limit commodity flows to satisfy the demands of all nodes.

Equation (4) shows the transformation of commodity flows, such as propellant burning, propellant boiloff, ISRU production, and crew consumables. For example, impulsive propellant consumption can be expressed as follows:

$$\begin{bmatrix} 1 & 0 & 0 \\ -\phi & 1-\phi & -\phi \\ 0 & 0 & 1 \end{bmatrix}_{vij} \begin{bmatrix} \text{payload} \\ \text{propellant} \\ \text{structure mass} \end{bmatrix}_{vijt}^+ = \begin{bmatrix} \text{payload} \\ \text{propellant} \\ \text{structure mass} \end{bmatrix}_{vijt}^- \quad (8)$$

In Eq. (8), ϕ is the propellant mass fraction that is defined by the rocket equation $\phi = 1 - \exp(-(\Delta V/I_{sp}g_0))$, where ΔV is the change in the vehicle's velocity along the arc, I_{sp} is the specific impulse, and g_0 is the standard gravity. Another example about propellant production from ISRU can be expressed as follows:

$$\begin{bmatrix} 1 & F_{\text{ISRU}}(\text{ISRU plant mass}) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \text{propellant} \\ \text{ISRU plant mass} \end{bmatrix}_{vijt}^+ = \begin{bmatrix} \text{propellant} \\ \text{ISRU plant mass} \end{bmatrix}_{vijt}^- \quad (9)$$

In Eq. (9), $F_{\text{ISRU}}(\text{ISRU plant mass})$ is the ISRU production rate. It is a function of the ISRU plant mass, which is introduced in detail in Sec. II.A.3.

Equation (5) is a spacecraft concurrency constraint, which corresponds to the upper bound of commodity flows limited by spacecraft propellant capacity or payload capacity. In this paper, we only consider spacecraft payload and propellant mass capacities as concurrency constraints. Therefore, the number of concurrency constraints is equal to two, which is indexed by l . For example, the concurrency constraints about payload and propellant can be expressed as follows:

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \text{payload} \\ \text{crew} \\ \text{consumables} \\ \text{propellant} \end{bmatrix}_{vijt}^+ \leq \begin{bmatrix} C_v \\ M_v \end{bmatrix} y_{vijt}^+ \quad (10)$$

where C_v and M_v are the payload and propellant capacity of spacecraft v . In past literature, Eq. (5) could also work as a linear spacecraft model through an inert mass fraction [2]. Note that there is an underlying assumption in this formulation that the spacecraft capabilities of the components can be additively combined. In reality, the interoperability between spacecraft can be significantly more complex.

Equation (6) is the time window of missions defined by time window matrix W_{ij} .

Equation (7) is spacecraft design model. It is introduced in detail next, in Sec. II.A.2.

The time index of this model is inspired by a biscale time-expanded network by Ho et al. [2]. In a biscale network model and nodes are partitioned into clusters, such that transportation across the same pair of clusters has a common time window [2]. The word "biscale" means a larger time-step scale is used among clusters and a smaller time-step scale is used at the intracluster level.

In this paper, based on the biscale time-expanded network, nonuniform time steps are used within clusters. If there are time windows within clusters, the time step length of the holdover arc is determined by the intervals among the time windows. For example, we have a network defined as shown in Fig. 3. Because the time windows of node k only open at time t , $t+2$, and $t+8$, there are only two holdover arcs for node k . The first one starts from time t and ends at $t+2$; the second one starts from time $t+2$ and ends at $t+8$. The time-step lengths of these holdover arcs are two and six in this example for node k . Nonuniform time steps can eliminate redundant time steps and increase computational efficiency.

2. Vehicle Model

This paper uses a data-based spacecraft model developed by Taylor in an integrated transportation system [15]. This is a nonlinear regression function based on preexisting spacecraft. The

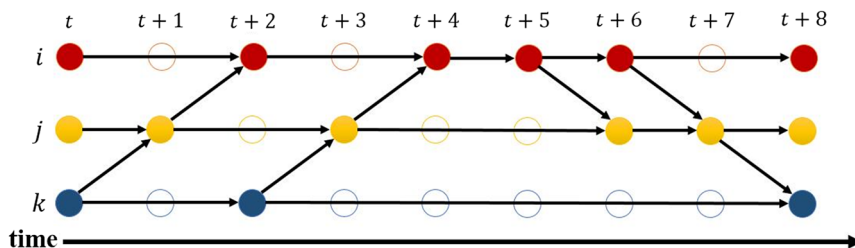


Fig. 3 Nonuniform time-step time-expanded network.

relationship among the structure mass and the three design variables is as follows [15]:

$$s^* = 2.3931 * C + \alpha(f) * M * \left(1 - \frac{0.2 * M}{M_{UB}}\right) + \frac{0.4189((M * I_{sp}(f) * g_0/t_b)^{0.7764}}{g_0} \quad (11)$$

where s^* is structure mass (or dry mass); C is the spacecraft payload capacity; M is the propellant capacity; f is the fuel type; g_0 is the gravitational acceleration on Earth, 9.8 m/s^2 ; t_b is the spacecraft impulsive burn time that is set as 120 s; M_{UB} is the upper bound allowed for the propellant tank capacity, which is assumed as 500,000 kg; and α and I_{sp} are the structural fraction and specific impulse that are determined by fuel type f . Note that α is the structure fraction of the spacecraft propellant tank defined in [15]. In this paper, only one kind of propellant liquid oxygen (LOX)/kerosene is considered in the spacecraft model. This is because, compared with other propellants, liquid oxygen (LOX)/liquid hydrogen (LH2), LOX/kerosene, and Monomethylhydrazine (MMH)/Nitrogen tetroxide (N2O4) have relatively high specific impulses I_{sp} with a relatively low structural fraction α (see Table A2). Considering the abundant availability of oxygen on the moon from ISRU and the large boiloff of hydrogen over the long campaign, LOX/kerosene is chosen.

The spacecraft data used to develop this spacecraft model are listed in the Appendix. A comparison of this spacecraft model with a conventional sizing method is shown in Table 2 [15,19–21]. We can find that, as a nonlinear regression model, this spacecraft sizing model has a higher fidelity as compared with a conventional method such as photographic scaling [5] when evaluating some historic spacecraft. However, limited by the accuracy of the spacecraft model, this nonlinear spacecraft model output can also diverge from historic spacecraft, such as the Apollo lunar module (LM) ascent stage (AS), the Apollo command module (CM), and the Apollo service module (SM), as shown in Table 2. Note that this paper does not intend to develop an accurate spacecraft model; instead, this paper proposes the methods for space logistics optimization with a nonlinear spacecraft model as an input. The aforementioned spacecraft model is chosen from the literature as an example to demonstrate our space logistics optimization method.

3. ISRU (Oxygen) Model

Only oxygen ISRU is considered in this paper because the oxidizer is a major component of spacecraft mass. Studies have shown that oxygen can be produced at a lower cost as compared with being

$$N_M = \begin{cases} 0 & \text{if } M_O \in [0, 400) \\ -0.438 + 6.9623 \left(1 - \exp\left(-\frac{M_O}{812.1563}\right)\right) + 2.0173 \left(1 - \exp\left(-\frac{M_O}{3967.2644}\right)\right) & \text{if } M_O \in [400, 10000) \end{cases} \quad (14)$$

delivered directly from Earth [1]. The Integrated ISRU system developed by Schreiner used a molten regolith electrolysis model and

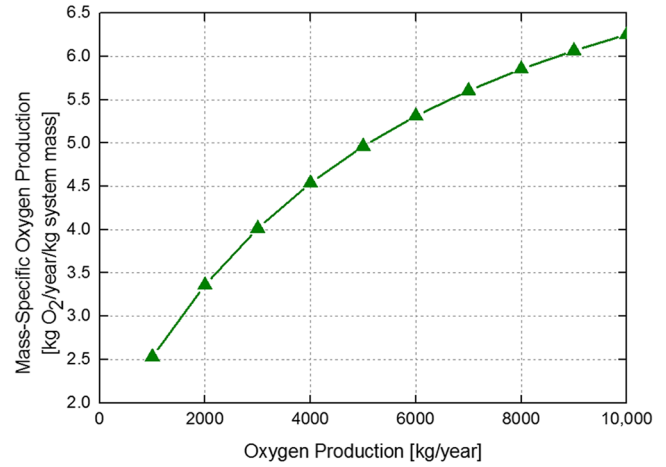


Fig. 4 Oxygen production level normalized by ISRU system mass [7].

balanced the tradeoff between optimal reactor performance and optimal excavator design. The ISRU system plant includes the reactor, Ytria-Stabilized Zirconia (YSZ) separator, excavator, hopper and feed system, oxygen liquefaction and storage system, and power system. For a detailed description of these systems, please see [7,22]. Because a separate cooler is designed to reliquefy oxygen that has boiled off in the storage system [7], the boiloff effect of oxygen produced by ISRU is not considered. According to Schreiner's research [7], the ISRU production rate is increasing with the ISRU plant mass. The relationship is shown in Fig. 4.

It can be expressed as the following function [7]:

$$N_M = 8.8 \exp\left(-\frac{4241}{2405 + N_O}\right) \quad (12)$$

where N_M is the oxygen production rate in kilograms of oxygen (O_2) per year per kilogram system mass, and N_O is the oxygen production per year. If we assume the ISRU plant mass as M_O , then

$$N_M = \frac{N_O}{M_O} \quad (13)$$

Equation (12) is a regression equation that has a horizontal asymptote as $N_O \rightarrow \infty$. Based on Eq. (12) and the data from Schreiner [7], we build another regression equation to get $N_M = F_{ISRU}(M_O)$, which is shown as follows:

Table 2 Spacecraft sizing comparison

Spacecraft	Payload capacity C , kg	Propellant capacity M , kg	Propellant	Actual structure mass s , kg	Nonlinear regression s^* , kg	Photographic scaling [5] s , kg
Centaur	0	20,830	LOX/LH2	2,462 [19]	3,131	3,215
Saturn V third stage	0	107,725	LOX/LH2	12,014 [15]	13,513	16,625
H-II transfer vehicle	6,000	2,000	MMH/N2O4	10,500 [20]	15,001	15,197
Automated transfer vehicle	5,500	2,613	MMH/N2O4	10,300 [21]	13,986	14,257
Apollo LM DS	500	8,804	N2O4/UDMH	2,770 [15]	2,505	2,566
Apollo LM AS	250	2358	N2O4/UDMH	1,719 [15]	1,005	687
Apollo CM	524	0	N2O4/UDMH	4,841 [15]	1,254	1,150
Apollo SM	60	18,413	N2O4/UDMH	6,053 [15]	2,682	5,367

The comparison between the regression equation [Eq. (14)] and Schreiner's model [7] is shown in Fig. 5. It is the relationship between

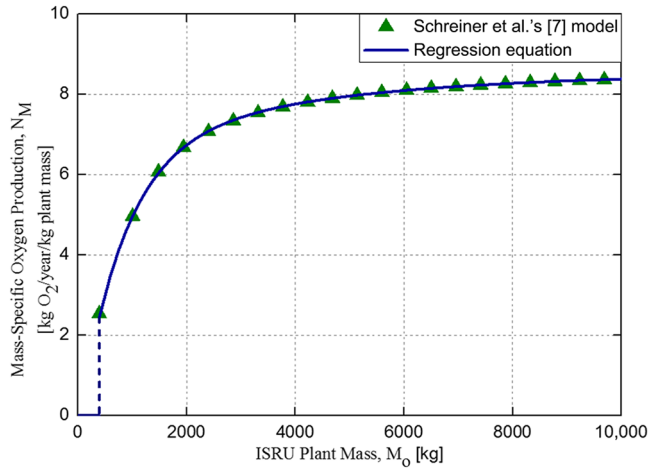


Fig. 5 Oxygen production normalized by ISRU system mass.

the ISRU system plant mass M_O and the oxygen production rate N_M . Figure 5 shows that Eq. (14) fits the data of Schreiner's model well. It is used to calculate oxygen production in our space logistics model.

Equation (14) is a piecewise nonlinear function. When $M_O < 400$, ISRU does not function. Because the ISRU system includes the reactor, excavator, storage system, power system, etc., if the total mass of the ISRU plant is too low, the integrated ISRU system does not contain enough components to start functioning as a unit. The domain of the ISRU plant mass M_O in this equation is from 0 to 10,000 kg. From Eqs. (13) and (14), we can get the relationship between the oxygen production every year and the total mass of the ISRU plant deployed, which is $N_O = M_O * F_{\text{ISRU}}(M_O)$.

B. Optimization Methods

The campaign-level space logistics problem, considering nonlinear spacecraft and the ISRU model, is a mixed-integer nonlinear programming (MINP) problem. This section introduces the piecewise function approximation method that linearizes the nonlinear ISRU and spacecraft models, and then converts them into binary mixed-integer programming formulations. Two approaches to solve this MINP problem are proposed. Both methods use the piecewise function approximation method for approximating the ISRU model. The difference between the two methods lies in how to deal with the nonlinear spacecraft model. The first approach converts the whole problem into a MILP formulation, which is expected to solve this problem more efficiently. The second approach is developed, based on simulated annealing. It separates the spacecraft design from the mission planning part, which is intended to create the ability to consider higher-fidelity spacecraft models if necessary.

1. Piecewise Function Approximation Method

The following shows the method to approximate a single variable nonlinear function by a piecewise linear function, and then convert it into a binary mixed-integer programming formulation. Considering a nonlinear function, a proper number of breakpoints inside the function domain is selected and then connected linearly, as shown in Fig. 6. This single variable nonlinear function is approximated by a piecewise linear function, which can be described as follows:

$$f(x) := \begin{cases} k_1x + c_1, & x \in [u_1, u_2] \\ k_2x + c_2, & x \in [u_2, u_3] \\ \vdots \\ k_{N-1}x + c_{N-1}, & x \in [u_{N-1}, u_N] \end{cases} \quad (15)$$

for some $N \in \mathbb{Z}_+$:

$$\{k_i\}_{i=1}^{N-1} \subset \mathbb{R}, \quad \{c_i\}_{i=1}^{N-1} \subset \mathbb{R}$$

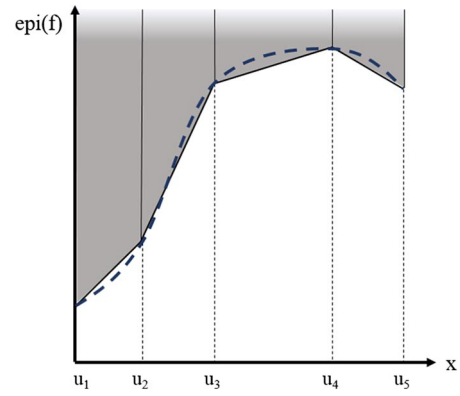


Fig. 6 Piecewise function and its epigraph as the union of polyhedra.

and

$$\{u_i\}_{i=1}^{N-1} \subset \mathbb{R}$$

where u_i is a breakpoint.

To solve this piecewise function in our mixed-integer linear programming model, the next step is modeling the piecewise function $f: D \subset \mathbb{R} \rightarrow \mathbb{R}$ as a binary mixed-integer programming formulation, where D is the domain of x . An appropriate approach to do it is to model its epigraph

$$\text{epi}(f) = \{(x, z) : x \in D, z \in \mathbb{R}, f(x) \leq z\}$$

as shown in Fig. 6. The epigraph of a piecewise function can be considered as a union of polyhedra $\mathcal{P}$. Each polyhedron P corresponds to an interval $[u_i, u_{i+1}]$. Therefore, we can rewrite the piecewise function f as follows:

$$f(x) := \{k_P x + c_P, x \in P, \quad \forall P \in \mathcal{P}\} \quad (16)$$

Vielma et al. [23] reviewed and introduced several ways to convert a union of polyhedra $\mathcal{P}$ into a binary mixed-integer programming formulation, including a convex combination model, logarithmic branching convex combination (referred to as log) model, a multiple choice model, an incremental method, etc. As one of the most performant formulations, the log model is chosen for piecewise linear approximation of spacecraft and ISRU models in this paper.

Following the theory described by Vielma et al. [23], $(x, z) \in \text{epi}(f)$ can be represented as the convex combination of points $(ve, f(ve))$ for $ve \in V(P)$, where V denotes the set of vertices (i.e., breakpoints) of polyhedron P , plus a ray in $C_1^+ := \{(0, z) \in \mathbb{R} \times \mathbb{R} : z \geq 0\}$. Then, a continuous variable λ is assumed for each vertex $ve \in V(P)$ and for each $P \in \mathcal{P}$. As a result, a point $(x, z) \in \text{epi}(f)$ can be represented as

$$(x, z) = r + \sum_{P \in \mathcal{P}} \sum_{ve \in V(P)} \lambda_{P,ve} (ve, f(ve))$$

for $r \in C_1^+$ and

$$\{\lambda_{P,ve}\}_{P \in \mathcal{P}, ve \in V(P)} \subset \mathbb{R}_+$$

such that

$$\sum_{P \in \mathcal{P}} \sum_{ve \in V(P)} \lambda_{P,ve} = 1$$

Moreover, we can further reduce the number of the continuous variable λ and the constraints of the formulation by identifying a binary branching scheme. As a result, we only need to define λ variables associated with the vertices (i.e., breakpoints) in union of polyhedra $\mathcal{P}$ rather than for each vertex in each polyhedron P . This can reduce the number of λ variables significantly because two adjacent polyhedra share the same vertex (i.e., breakpoint).

Finally, the resulting formulation is given by

$$\sum_{ve \in \mathcal{V}(\mathcal{P})} \lambda_{ve} ve = x, \quad \sum_{ve \in \mathcal{V}(\mathcal{P})} \lambda_{ve} (k_P ve + c_P) \leq z \quad (17a)$$

$$\lambda_{ve} \geq 0 \quad \forall ve \in \mathcal{V}(\mathcal{P}), \quad \sum_{ve \in \mathcal{V}(\mathcal{P})} \lambda_{ve} = 1 \quad (17b)$$

$$\sum_{ve \in L_u} \lambda_{ve} \leq y_u, \quad \sum_{ve \in R_u} \lambda_{ve} \leq (1 - y_u), \quad y_u \in \{0, 1\} \quad \forall u \in U \quad (17c)$$

where $\{L_u, R_u\}_{u \in U}$ is a family of dichotomies, which is a branching scheme of λ variables. This formulation is called the logarithmic branching convex combination model. The branching scheme was introduced by Vielma and Nemhauser [24], and details of the log method can be found in [23]. Using the aforementioned method, an example of building a log formulation is shown in the following.

For example, the nonlinear function in Fig. 6 can be approximated by a piecewise function using five breakpoints given by

$$f(x) := \begin{cases} k_1 x + c_1, & x \in [u_1, u_2] \\ k_2 x + c_2, & x \in [u_2, u_3] \\ k_3 x + c_3, & x \in [u_3, u_4] \\ k_4 x + c_4, & x \in [u_4, u_5] \end{cases} \quad (18)$$

Then, a point (x, z) in this piecewise function can be expressed as a binary mixed-integer programming formulation using the log formulation given by

$$u_1 \lambda_1 + u_2 \lambda_2 + u_3 \lambda_3 + u_4 \lambda_4 + u_5 \lambda_5 = x \quad (19a)$$

$$f(u_1) \lambda_1 + f(u_2) \lambda_2 + f(u_3) \lambda_3 + f(u_4) \lambda_4 + f(u_5) \lambda_5 \leq z \quad (19b)$$

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 = 1 \quad \forall \lambda_i \geq 0, \quad i \in \{1, \dots, 5\} \quad (19c)$$

$$\lambda_1 + \lambda_5 \leq y_1, \quad \lambda_3 \leq 1 - y_1, \quad \lambda_1 + \lambda_2 \leq y_2, \quad \lambda_4 + \lambda_5 \leq 1 - y_2, \quad y_1, y_2 \in \{0, 1\} \quad (19d)$$

2. Mixed-Integer Linear Programming Optimization Framework

From the MILP formulation [Eqs. (1–7)], we can see that, if the spacecraft model is considered together with mission planning, even though it is approximated using a piecewise function and then converted into a binary mixed-integer programming formulation, there are still quadratic terms. These quadratic terms exist in an objective function [Eq. (1)], a commodity transformation constraint [Eq. (4)], and a concurrency constraint [Eq. (5)]. The reason is that spacecraft design variables s_v , C_v , and M_v are also variables in the MILP formulation. These quadratic terms are the products of discrete and continuous variables. Because the coefficient matrix of quadratic terms is not positive semidefinite, to solve this model, linearizing quadratic terms is a better choice.

Consider a nonlinear equation $z_{vijkt} = O_v b_{vijkt}$, where O_v is a continuous variable representing the dry mass s_v , payload capacity C_v , or propellant capacity M_v of spacecraft v ; and b_{vijkt} is a binary variable representing whether the k th spacecraft is flying along the arc from i to j at time t . Note that v represents the type of spacecraft, and there are multiple spacecraft for the same type. Also, k is the index of spacecraft of the same type. Assuming the maximum number of spacecraft v that can be used is n , we have

$$y_{vijt} = \sum_{k=1}^n b_{vijkt} \quad (20)$$

Therefore, using the big- $\mathcal{M}$ method, product $z_{vijkt} = O_v b_{vijkt}$ can be linearized as follows:

$$z_{vijkt} \leq M b_{vijkt} \quad (21a)$$

$$z_{vijkt} \leq O_v \quad (21b)$$

$$z_{vijkt} \geq O_v - (1 - b_{vijkt})M \quad (21c)$$

$$z_{vijkt} \geq 0 \quad (21d)$$

where M is a large constant.

As a result, after approximating the nonlinear model of ISRU and spacecraft by a piecewise function approximation method and linearizing quadratic terms in concurrency constraints, all nonlinear terms are converted into linear constraints. The campaign-level space logistics optimization problem considering nonlinear spacecraft and ISRU production models finally becomes a mixed-integer linear programming problem.

3. Simulated Annealing Optimization Framework

SA is a conventional heuristic optimization method that cannot guarantee or certify its global optimality. Typically, it takes a long time to achieve an acceptable solution. This SA-based method is inspired by the optimization framework developed by Taylor [15]. However, Taylor's approach cannot take into account infrastructure such as ISRU because the mission planning part of the framework uses a shortest-path-based approach. This paper improves this method by using an arc-based formulation and solves it as a multicommodity network flow problem.

The only difference between the MILP-based method and the SA-based method is how to deal with the nonlinear spacecraft model. The SA-based approach separates the spacecraft model from mission planning. As a result, the spacecraft design variables C_v and M_v are constant inside the MILP formulation. They are only variables in the separated spacecraft model. In other words, the quadratic terms in the MILP formulation disappear. The nonlinear ISRU model is still designed concurrently with the mission planning part. It is converted into a binary mixed-integer programming formulation through piecewise linear approximation.

Because the spacecraft model is separated, the advantage of the SA approach is that it provides the possibility of extending the spacecraft model in an easier manner when compared to the method of MILP formulation.

A flowchart of SA is shown in Fig. 7. The mission planning section is actually the evaluation step of simulated annealing. After selecting random neighbors in the spacecraft model, the performances of the spacecraft are evaluated by a space logistics mission planning optimization using these spacecraft. The best space logistics solution is recorded. The optimization algorithm repeats until a stopping criteria (e.g., convergence, computation budget), and then optimal spacecraft parameters and space logistics solutions are output.

III. Results and Analysis

A. Model Validation

This section validates the network model using the Apollo 17 mission as an example, where no ISRU and spacecraft design models are considered. Instead, the original spacecraft in the Apollo mission is used in mission planning. The Apollo 17 mission can be modeled as a four-node network, as shown in Fig. 8, including the Pacific Ocean (PAC), low Earth orbit (LEO), low lunar orbit (LLO), and lunar surface (LS). The time of flight (TOF) and ΔV for each transportation arc are also shown in Fig. 8. We assume that one day is one time step.

Note that the propellant cost for transportation from Earth to LEO is not considered because the IMLEO is minimized in this paper. For a transportation system, the IMLEO is a widely accepted measure of the mission cost [1,3]. In some literature, the financial cost is also used as a metric [5]. However, this paper focuses on limited resource utilization and the interaction between the mission planning and spacecraft design. To simplify the problem, the IMLEO is used as a metric for the mission cost in this paper.

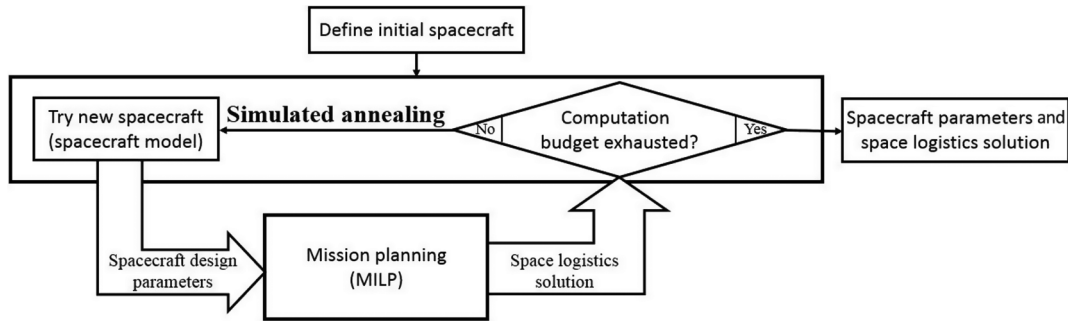


Fig. 7 Flowchart of SA-based approach.

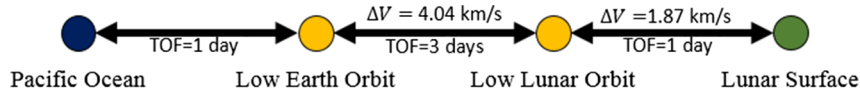


Fig. 8 Apollo 17 network model.

Table 3 lists all spacecraft used in Apollo 17, and Table 4 lists the demand and supply of the Apollo 17 mission. In this example, two astronauts, along with the scientific experiment equipment, are sent to the lunar surface and stay there for one day; whereas a third astronaut stays in lunar orbit for three days. Then, all three astronauts come back with 110 kg of lunar samples. For this Apollo 17 mission, the time window is always open.

The optimized solution using the GMCNF model for this example is depicted in Fig. 9. All commodities are delivered together from LEO. Notice that crew members fly back to the Earth by LM descent stage (LMDS) instead of the Apollo command/service module (CSM) in the original mission. This is a feasible solution because the payload capacity of the LMDS is large enough to contain three crew members and 110 kg of lunar samples. The size of the propellant tank is also sufficient to support the return directly from lunar orbit. This is also an optimal solution because one important function of the CSM is to bring crew members and lunar samples back to the Earth. If the LMDS can complete this work, the service module would not be used because the LMDS

has a smaller structure mass, which means less propellant cost. A similar result is observed in Taylor et al.'s study [3]. Saturn V's third stage performs as a propellant depot after reaching the lunar orbit. The detailed mass flows are shown in Fig. 10. The consumables in Fig. 10 include water, oxygen, and food for the crew.

Table 5 compares the differences of the payload and propellant at Earth orbit insertion between the optimized Apollo 17 mission and the original mission [25]. Notice that no spacecraft design model is considered in this example. The optimized Apollo mission uses the same vehicles as original mission.

The GMCNF model actually selects a better solution for the Apollo 17 mission. For validation purposes, we can manually track what has changed as compared with the original mission. Starting from the original Apollo 17 mission, we remove the service module, the propellant in LM ascent stage (LMAS), and the propellant to transport these commodities to lunar orbit. Then, we add the propellant for the LMDS to support its return to Earth. Finally, the initial mass in LEO becomes 105,614 kg. The optimal solution by the GMCNF model is 105,750 kg, which is only a 0.1% difference from our manual validation results. Thus, we can conclude that the time-expanded GMCNF could find an optimal solution and has sufficient accuracy to optimize the space logistic problem.

Table 3 Spacecraft used in Apollo 17

Spacecraft	Propellant capacity M , kg	I_{sp} , s	Payload capacity C , kg	Structure mass s , kg
Saturn V second stage	452,045	421	0	38,415
Saturn V third stage	107,725	421	0	12,014
Command module	0	0	524	4,841
Service module	18,413	314	60	6,053
LM descent stage	8,804	311	500	2,770
LM ascent stage	2,358	311	250	1,719

Table 4 Demand and supply of Apollo 17 mission

Payload type	Node	Demand time, day	Supply
Crew, ^a no.	Lunar surface	5	-2
Crew, ^a no.	Lunar orbit	4	-1
Crew return, ^a no.	Lunar surface	6	2
Crew return, ^a no.	Lunar orbit	7	1
Crew return, ^a no.	Earth	11	-3
Equipment, ^b kg	Lunar surface	5	-420
Sample, ^c kg	Earth	11	-110
Crew, equipment, ^b kg	Earth	0-11	+∞
Sample, ^c kg	Lunar surface	0-11	+∞

^aCrew and crew returning are separated in order to ensure that the crew shall reach the moon before returning to Earth.

^bEquipment denotes all scientific experiment equipment for lunar surface activities, including a lunar roving vehicle.

^cSample denotes the commodities we want to bring back from space, typically for scientific research.

B. Campaign-Level Space Logistics Design

This section evaluates and analyzes the performance of the space logistics optimization framework for mission planning and spacecraft design. Both ISRU and spacecraft design models are taken into account. Moreover, maintenance costs and the oxygen boiloff are also considered in this case. This is a MILP problem solved in Python by a Gurobi 6.5 solver on an Intel Core i7-4700MQ quad-core 3.4 GHz platform. All the following numerical experiments in this paper are solved on this platform. The piecewise linear approximations for nonlinear ISRU and spacecraft models are done through a Python-based open-source optimization package named Pyomo.

In this section, we first perform an optimization for a single space mission as the baseline solution. Then, we optimize a campaign, which contains multiple space missions, and compare its result with the baseline solution to study the effect of campaign-level space logistics design. These space mission optimizations are all completed through the MILP-based method. Next, we compare the performance of the MILP-based method with the SA-based method. Finally, we perform sensitivity analyses of ISRU productivity and spacecraft design.

1. Single Mission Design with MILP

In this part, we optimize a single space mission as the baseline solution. It is used to demonstrate the effect of campaign-level space logistics design in the next section. Based on the Apollo 17 mission, a new large-scale lunar mission is designed as an example for

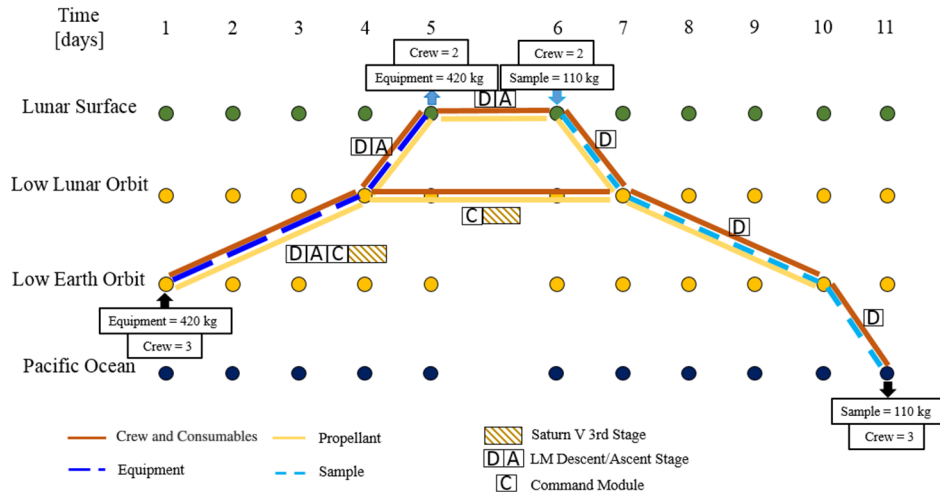


Fig. 9 Apollo 17 example.

From	LEO	LLO	LLO	LS	LS	LLO	LEO
To	LLO	LS	LLO	LS	LLO	LEO	PAC
Time, days	1-4	4-5	4-9	5-8	8-9	9-12	12-13
Saturn V third stage, no.	1	—	1	—	—	—	—
Command module, no.	1	—	1	—	—	—	—
LM descent stage, no.	1	1	—	1	1	1	1
LM ascent stage, no.	1	1	—	1	—	—	—
Crew, no.	3	2	1	2	2	3	3
Consumables, kg	124	26	51	17	9	51	13
Equipment, kg	420	420	—	—	—	—	—
Sample, kg	—	—	—	—	110	110	110
Propellant, kg	83,562	8,940	8,656	2,531	2,531	8,656	—
Total Flow, kg	105,750	14,075	25,662	7,237	5,620	11,887	3,193

Fig. 10 Commodity flows in optimal solution.

optimization. The demand and supply are shown in Table 6. In this case, 12 crews are delivered to the moon with 4200 kg of equipment and come back with 500 kg samples. The assumptions are listed in Table 7 [26,27]. For this single mission example, the time window is always open.

The percentage of oxygen in propellant in Table 7 is used to consider the production of the oxygen ISRU plant and the boiloff effect. Note that the oxygen consumed by crew members is very low as compared with the oxidizer in propellant (LOX/kerosene). To simplify the problem, the oxygen produced by the ISRU plant is only used for the propellant oxidizer.

The maintenance of ISRU is performed in every mission, and the amount of ISRU maintenance mass is determined by the ISRU plant mass. In addition, the maintenance of spacecraft is performed in every flight and the amount of needed maintenance mass is related to the spacecraft structure mass.

The solution of the single mission optimization is listed as follows. Two types of spacecraft are considered, and two spacecraft can be used for each type.

Table 8 lists the spacecraft design in this case, and the mission planning solution is depicted in Fig. 11. Because the mission only lasts for 13 days, no ISRU plant is deployed on the moon. This solution is

Table 5 Mass differences comparison at Earth orbit insertion

Spacecraft	Class of mass	Original Apollo 17, kg	Optimized Apollo 17, kg
Saturn V third stage	Propellant	96,204	74,758
	Structure mass	12,014	12,014
Command module	Crew member	3*100	—
	Consumables	124	—
	Equipment	—	420
	Structure mass	4,841	4,841
Service module	Propellant	18,413	—
	Structure mass	6,110	—
LM descent stage	Crew member	—	3*100
	Consumables	—	124
	Equipment	420	—
	Propellant	8,804	8,804
LM ascent stage	Structure mass	2,770	2,770
	Propellant	2,358	—
All spacecraft	Structure mass	1,719	1,719
	Total mass	154,077	105,750

Table 6 Demand and supply of lunar mission

Payload type	Node	Demand time, day	Supply
Crew, no.	Lunar surface	5	−12
Crew return, no.	Lunar surface	8	12
Crew return, no.	Earth	13	−12
Equipment, kg	Lunar surface	5	−4200
Sample, kg	Earth	13	−500
Crew, equipment, kg	Earth	0–13	+∞
Sample, kg	Lunar surface	0–13	+∞

Table 7 Summary of parameters and assumptions

Parameter	Assumed value
Percentage of oxygen in propellant	71.91% LOX/kerosene
Oxygen boiloff rate	0.016%/day
Food consumption rate [26,27]	1.015 kg/day/crew
Water consumption rate [27]	6.37 kg/day/crew
Oxygen consumption rate [27]	1.18 kg/day/crew
Crew mass (including space suit)	100 kg/crew
ISRU maintenance	10%/year
Spacecraft maintenance	1%/flight

Table 8 Spacecraft design of the single mission design solution

Spacecraft type	Payload capacity C, kg	Propellant capacity M, kg	Structure mass S, kg
1	2,020	166,481	17,996
2	2,262	23,891	7,342

also an optimal solution of the conventional carry-along strategy. All propellant and consumables used in the mission come from the Earth. The spacecraft size is also shown in Fig. 11, corresponding to the

structure mass in Table 8. Even though the spacecraft are designed to be multiuse without specific functions assigned for each type, they still perform different roles in the mission. Spacecraft 1, which is relatively larger, is mainly used to deliver equipment. Spacecraft 2, which is smaller, is mainly used to deliver crews and samples back from the moon. The detailed mass flows are shown in Fig. 12, and we can see that the IMLEO of this mission is 439,375 kg.

2. Campaign-Level Mission Design with MILP

This section studies the effect of campaign-level space logistics mission design. Three lunar missions are combined together into a campaign. In each mission, 12 crews are delivered to the moon with 4200 kg of equipment and come back to Earth with 500 kg of lunar samples. All the assumptions remain the same as before. For a multi-mission campaign, the time windows are open at the start of each mission, which are determined by the mission intervals. It takes about 2000 s to optimize this problem. The solution of campaign-level mission optimization with one year mission intervals is shown in the following:

Table 9 lists the spacecraft design of this case. The mission planning of this campaign is depicted in Fig. 13. We can see that the size of spacecraft 2 is the same as it in the single mission case, as shown in Table 8. The reason is that, like the single mission case, spacecraft 2 is mainly used to deliver crews and samples. Because the demands of crews and samples do not change, the size of their spacecraft 2 does not change. However, the size of spacecraft 1 is larger because, in a multi-mission campaign, spacecraft 1 is used to deliver not only the equipment but also ISRU plants.

The detailed mass flows are listed in Fig. 14. From Fig. 14, we can see that the IMLEO for each mission is decreasing. The first mission has the highest IMLEO because the ISRU plant is deployed in this mission. Starting from the second mission, the oxidizer used in the spacecraft flying out of the moon is produced by the ISRU plant. It

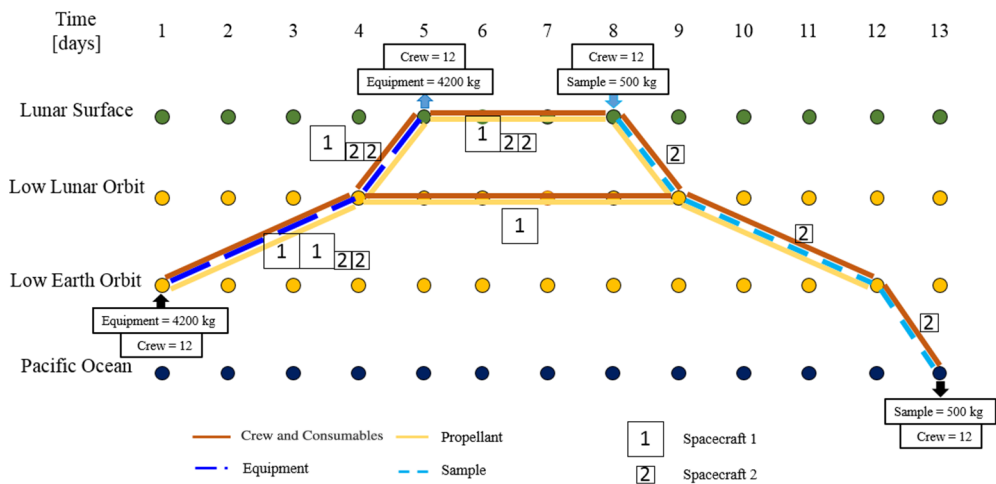


Fig. 11 Single lunar mission solution.

From	LEO	LLO	LLO	LS	LS	LLO	LEO
To	LLO	LS	LLO	LS	LLO	LEO	PAC
Time, days	1-4	4-5	4-9	5-8	8-9	9-12	12-13
Spacecraft 1, no.	2	1	1	1	—	—	—
Spacecraft 2, no.	2	2	—	2	1	1	1
Crew, no.	12	12	—	—	12	12	12
Consumables, kg	1,247	513	415	415	104	415	104
Maintenance, kg	1,307	474	327	147	73	147	73
Equipment, kg	4,200	4,200	—	—	—	—	—
Sample, kg	—	—	—	—	500	500	500
Oxidizer, kg	273,795	31,251	17,183	5,191	5,190	17,180	—
Kerosene, kg	106,950	12,207	6,711	2,027	2,027	6,711	—
Total Flow, kg	439,375	82,525	42,632	40,460	16,436	33,495	9,219

Fig. 12 Commodity flows of the single lunar mission solution.

Table 9 Spacecraft design of the multi-mission campaign design solution (interval: one year)

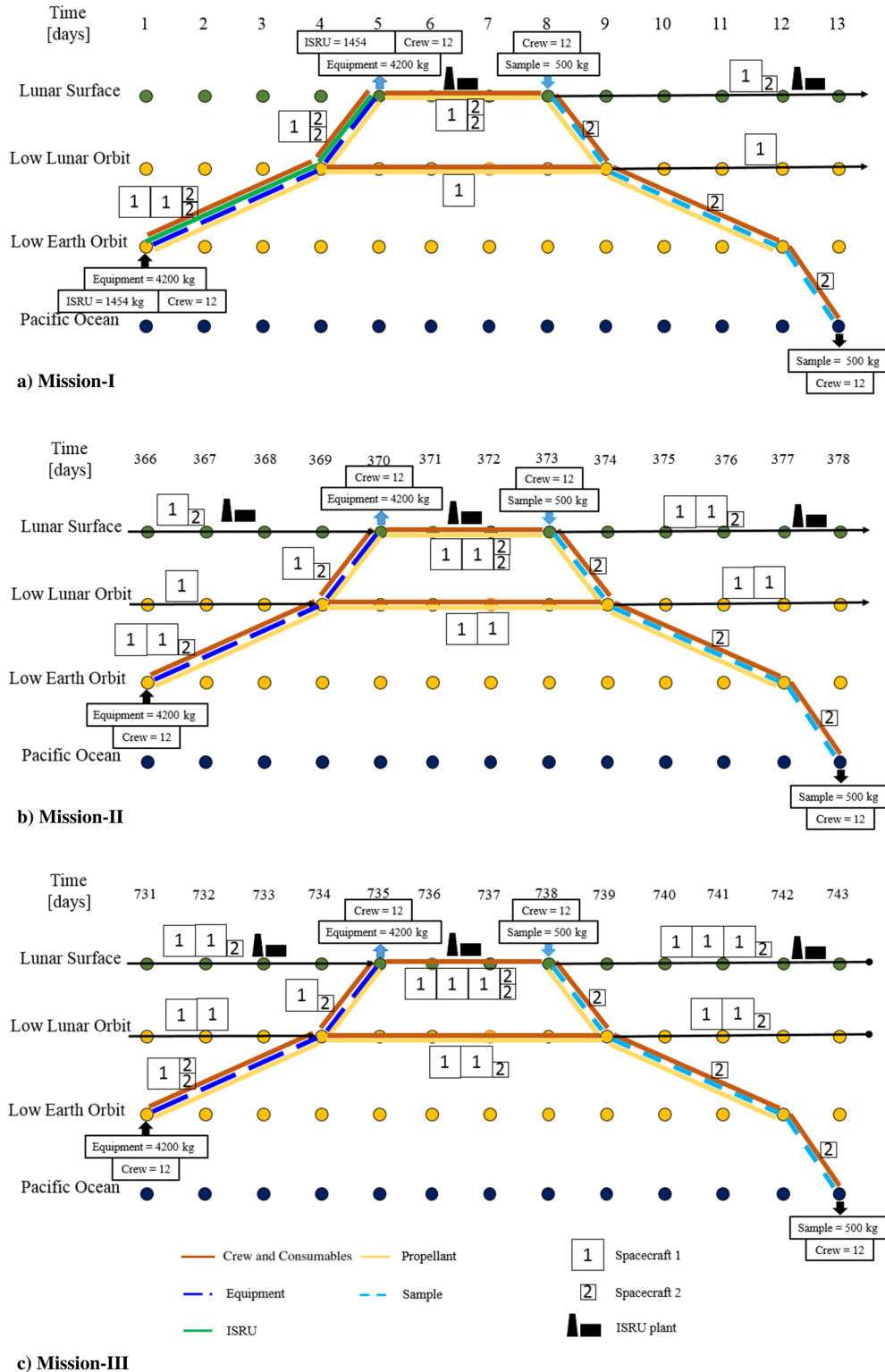
Spacecraft type	Payload capacity C , kg	Propellant capacity M , kg	Structure mass S , kg
1	3,978	198,580	24,745
2	2,262	23,891	7,342

relies less on the Earth resource for propellant oxidizer, which is expensive to deliver from Earth. As a result, the total IMLEO of this multi-mission campaign is lower than three single missions, as can be

seen by comparing Figs. 12 and 14. If this mission is repeated in the fourth year, because there is no need to deploy ISRU plants again, the mission cost is still lower than a single lunar mission.

The mission time intervals determine the total working time of the ISRU plant during the campaign. To study the effect of mission interval, the spacecraft structure mass and mission planning results with respect to the mission interval are depicted in Fig. 15.

As shown in Fig. 15, the IMLEO first decreases quickly because the mission obtains lots of benefits from the oxygen ISRU plant with a longer mission interval. However, after the mission interval is longer than 10 years, the IMLEO starts to decrease very slowly with

**Fig. 13** Multi-mission lunar campaign solution (interval: one year).

From	LEO	LLO	LLO	LS	LS	LLO	LEO
To	LLO	LS	LLO	LS	LLO	LEO	PAC
Mission-I							
Time, days	1-4	4-5	4-9	5-8	8-9	9-12	12-13
Spacecraft 1, no.	2	1	1	1	—	—	—
Spacecraft 2, no.	2	2	—	2	1	1	1
Crew, no.	12	12	—	12	12	12	12
Consumables, kg	1,247	519	415	415	104	415	104
Maintenance, kg	2,165	1,129	394	735	73	147	73
ISRU, kg	1,454	1,454	—	1,454	—	—	—
Equipment, kg	4,200	4,200	—	4,200	—	—	—
Sample, kg	—	—	—	—	500	500	500
Oxidizer, kg	319,854	36,273	17,183	5,165	5,165+25	1,7180	—
Kerosene, kg	125,087	14,315	6,711	2,163	2,027	6,711	—
Total Flow, kg	519,381	98,519	49,448	54,761	16,436	33,495	9,219
Mission-II							
Time, days	366-369	369-370	369-374	370-373	373-374	374-377	377-378
Spacecraft 1, no.	2	1	2	2	—	—	—
Spacecraft 2, no.	1	1	—	2	1	1	1
Crew, no.	12	12	—	12	12	12	12
Consumables, kg	1,246	519	415	415	104	415	104
Maintenance, kg	2,072	320	1,183	440	73	147	73
ISRU, kg	—	—	—	1,454	—	—	—
Equipment, kg	4,200	4,200	—	8,400	—	—	—
Sample, kg	—	—	—	—	500	500	500
Oxidizer, kg	287,032	22,995	14,453	9,432	9,456	14,395+2,785	—
Kerosene, kg	134,018	11,501	25,023	2,654	2,628	6,711	—
Total Flow, kg	486,600	72,822	90,564	88,169	21,303	33,495	9,219
Mission-III							
Time, days	731-734	734-735	734-739	735-738	738-739	739-742	742-743
Spacecraft 1, no.	1	1	2	3	—	—	—
Spacecraft 2, no.	2	1	1	2	1	1	1
Crew, no.	12	12	—	12	12	12	12
Consumables, kg	1,247	519	415	415	104	415	104
Maintenance, kg	394	320	220	147	73	147	73
ISRU, kg	—	—	—	1,454	—	—	—
Equipment, kg	4,200	4,200	—	12,600	—	—	—
Sample, kg	—	—	—	—	500	500	500
Oxidizer, kg	187,691	23,041	14,453	9,432	9,456	14,395+2,785	—
Kerosene, kg	58,671	11,601	6,711	2,628	2,628	6,711	—
Total Flow, kg	292,832	72,968	78,631	116,795	21,303	33,495	9,219

“+” : this part of propellant oxidizer is produced by ISRU plant

Fig. 14 Commodity flows of the multi-mission lunar campaign solution (interval one year).

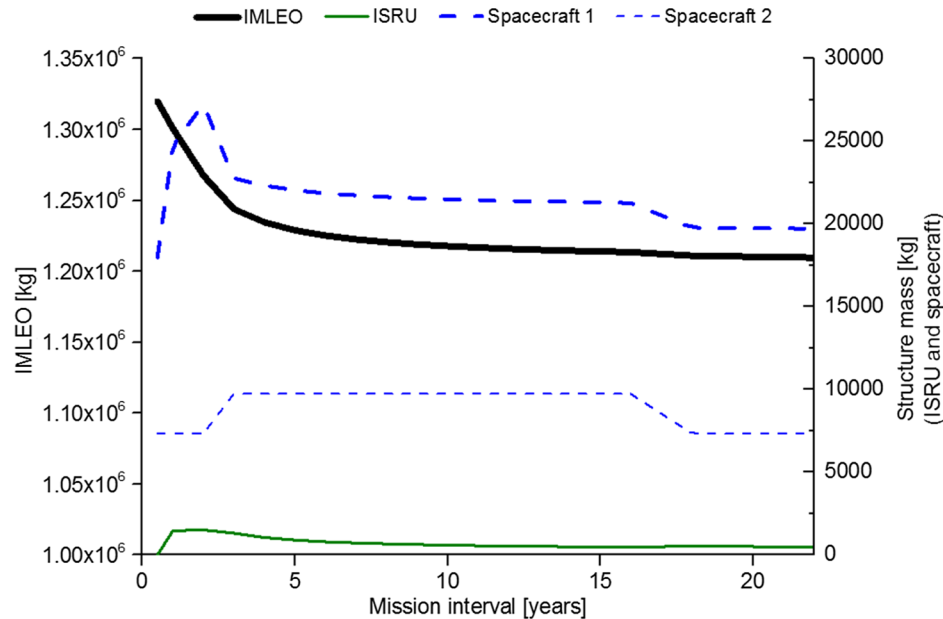


Fig. 15 Results of multi-mission lunar campaign design with respect to mission interval.

the increase of mission interval because the oxygen ISRU can only produce propellant oxidizer. No matter how much oxidizer can be produced by the ISRU plant, we still need kerosene delivered from Earth. When the mission interval is long enough, the benefit from the oxygen ISRU plant diminishes.

Furthermore, two types of spacecraft are considered. There is a direct relationship between the size of a large spacecraft (e.g., spacecraft 1) and the mass of the ISRU plant deployed. In Fig. 15, the variation pattern of the structure mass of spacecraft 1 matches the variation pattern of the ISRU plant mass. Similar to the results in campaign-level mission design, the large spacecraft is mainly used to deliver equipment and the ISRU plants. Because the demand of equipment is constant in each mission, the size of a large spacecraft is mainly determined by the ISRU plants; however, the small spacecraft is mainly used to deliver samples and crew members. Even if all spacecraft are designed in a general way (i.e., no specified function identified at the start of spacecraft design), the solver identifies the functions for each type of spacecraft.

3. Optimization Method Comparison

As discussed in the Methodology section (Sec. II), there are two optimization methods proposed in this paper: the MILP method and the SA optimization framework. This section studies the influences of different optimization methods. These methods are used to solve the same campaign-level mission design problem defined previously.

In the previous section, the nonlinear ISRU model and nonlinear spacecraft model are finally converted into a binary mixed-integer programming formulation through piecewise functions. However, these models can initially be assumed as linear to simplify the problem. For the linear ISRU model, we assume that the oxygen production rate is 7.5 kg of O₂ per year per kilogram of ISRU system mass. The linear spacecraft model is a linear simplification of Eq. (11), which is assumed as

$$S = 2.3931C + 0.09M \quad (22)$$

The comparison of different optimization methods is shown in Table 10.

The IMLEO of the problem with the linear spacecraft model is much higher when compared with the normal case (i.e., all nonlinear). This is because, in the nonlinear spacecraft model, the structure mass ratio of a large spacecraft is lower than a small spacecraft. The structure mass increases linearly with the spacecraft size in Eq. (22), which is developed based on small spacecraft; thus, the structure mass of a large spacecraft based on Eq. (22) is larger than reality. More propellant is consumed due to the large structure mass. On the contrary, the problem with the linear ISRU model has a lower IMLEO as compared with a nominal scenario. The reason is that the production rate of ISRU is increasing with the ISRU plant mass. As shown in Fig. 5, the ISRU would not work before the system mass reaches 400 kg. Then, the production rate is only 2.5 kg of O₂ per year per kilogram of ISRU system mass, with a 400 kg ISRU plant and 5 kg of O₂ per year per kilogram of ISRU system mass with a 1000 kg plant. To achieve a production rate of 7.5 kg of O₂ per year per kilogram of ISRU system mass, more than 3000 kg of ISRU plant should be deployed. Thus, the ISRU

production rate of a linear ISRU model is typically higher than the actual case.

The optimization time cost listed in Table 10 is the running time of the Gurobi solver, ignoring the model input time, which is typically 1 or 2 min. Because a MILP problem is generally solved by a linear-programming-based branch-and-bound algorithm, the calculation time is influenced by multiple factors, such as mission interval and ISRU or spacecraft model. For this four-node network in the space logistics model, considering a campaign including three lunar missions, there are approximately 100,000 constraints, 1500 integer variables, and 5400 continuous variables. In Table 10, the optimization time of all linear MILP methods is 1360 s when the mission intervals are one year. In fact, it may take a longer time when the mission intervals change. When we compare this all-linear MILP method with our method based on a piecewise linear function and MILP, we can find that our methods solve the space logistics problem including nonlinear ISRU and spacecraft models without much deterioration of computational efficiency. Our method may bring in several binary variables and constraints, but most of the constraints and variables come from the mission planning part, which is a GMCNF model. These constraints and variables are not influenced by which ISRU or spacecraft model is considered. Therefore, no matter which ISRU or spacecraft model is considered, the optimization time cost of this campaign-level mission design solved by the MILP method is at the same order of magnitude of approximately 1500–2500 s.

The same problem is also solved by the SA-based method. It cannot guarantee an optimal solution, but it is a good method if the precise global optimum is less important and an acceptable local optimum is wanted in a given time horizon. The biggest advantage of SA in this problem is that the spacecraft model is separated from the mission planning part. As a result, higher-fidelity spacecraft models can easily be considered, such as considering the fuel types. Moreover, different spacecraft can easily be assigned different functions initially in a separate spacecraft model.

The solution of SA is influenced significantly by the cooling strategy and the initial value of the problem (e.g., initial spacecraft). The start temperature is set as 10,000 and the algorithm will stop after the temperature reaches 0.1. The annealing schedule is an exponential cooling scheme with a cooling rate assumed as 0.985. Thus, there are 762 iterations in total. The initial spacecraft in this paper is generated by the spacecraft model randomly. Therefore, the final solution of the SA method may vary even for the same scenario.

The comparison of IMLEO between the MILP method and the SA method in different mission intervals is shown in Fig. 16. Because finding the optimal strategy to minimize IMLEO in space logistics is a minimization problem and the SA method always achieves a feasible (and suboptimal) solution, we can observe that the results from SA are typically 8–15% higher than the results of the MILP method. The gap is dependent on the candidate spacecraft generation method and cooling strategy.

We can improve the performance of the SA method by starting from a better initial solution with heuristics or optimizing the cooling strategy. Because the spacecraft model is separated from the mission planning part, SA is a better choice if a higher-fidelity spacecraft model needs to be considered and the precise global optimum is less important than finding an acceptable local optimum within given time.

4. Sensitivity Analysis

With the development of technology, ISRU could have a higher production rate and the spacecraft could have a lighter structure while maintaining sufficient strength at the same time. On the other side, some uncertainty factors may prevent the ISRU system from working properly or the spacecraft may be assigned extra structure mass. This section studies the sensitivity of this campaign-level mission design to the ISRU productivity and the spacecraft structure mass using the MILP method.

The considered problem is the campaign-level mission design with three lunar missions under the same assumptions as before. The mission interval is one year. The IMLEO with respect to ISRU

Table 10 Comparison of different optimization methods (interval: one year)

Optimization method		IMLEO, kg	Optimization time, s
MILP	ISRU model		
	Spacecraft model		
	Nonlinear	1,298,813	2,000
	Nonlinear	1,424,484	1,705
	Linear	1,296,999	2,566
	Linear	1,424,484	1,360
Simulated annealing		≈1,407,000	≈4,000

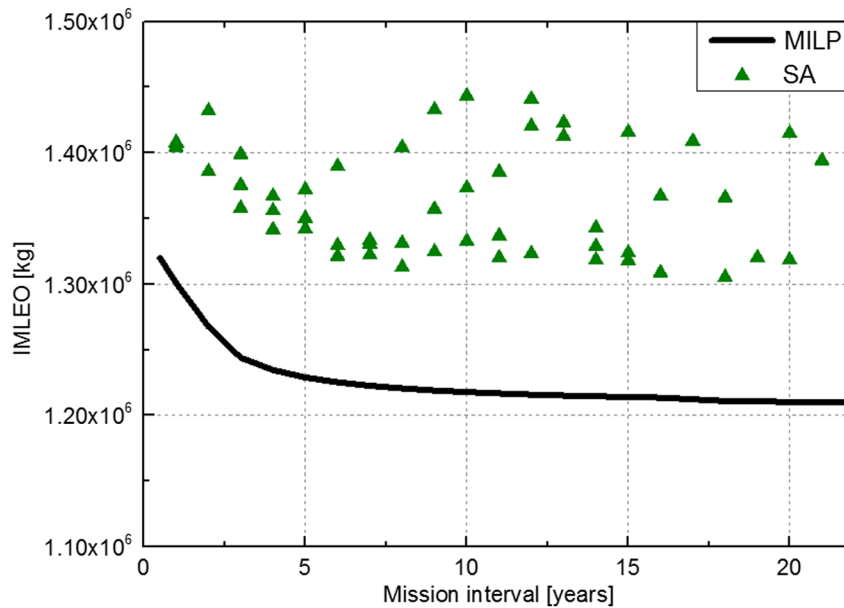


Fig. 16 Comparison of IMLEO results by MILP and SA method.

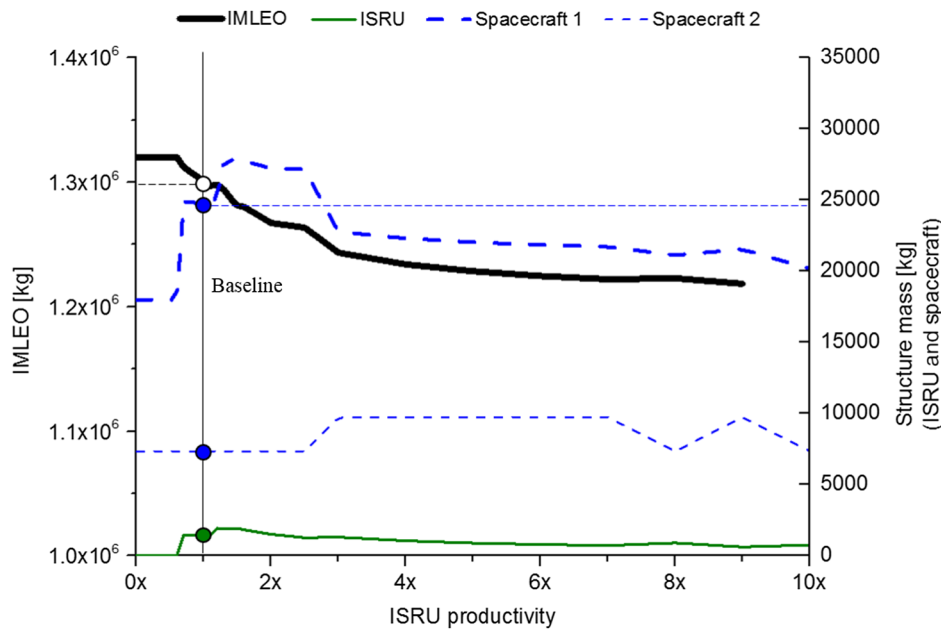


Fig. 17 Results of multi-mission lunar campaign design with respect to ISRU productivity (interval: one year).

productivity is shown in Fig. 17. The baseline is the normal ISRU production rate in Eq. (14).

As shown in Fig. 17, with the increase of ISRU productivity, the IMLEO of space mission is decreasing. When the oxygen ISRU productivity is less than 80% of the normal rate, no ISRU plant is deployed. The IMLEO and spacecraft design stay constant until the productivity reaches 80%. Then, with the increase of ISRU productivity, the IMLEO of the whole mission decreases monotonically. When the ISRU productivity reaches 150% of the normal production rate, the mass of the ISRU plant deployed starts to decrease slowly. The structure mass of the large spacecraft (i.e., spacecraft 1) still matches the variation pattern of the ISRU plant mass deployed in this case. Although the structure mass of the small spacecraft (e.g., spacecraft 2) varies in a certain range. One important note is that the increase of ISRU production does not reduce IMLEO unlimitedly. When the ISRU productivity is high enough, the mission

cost sensitivity to ISRU productivity is very low. With the increase of ISRU productivity, the total IMLEO is decreasing slowly. The reason is that the oxygen ISRU plant only produces oxidizer for this mission. The transportation still needs kerosene supplied from Earth.

If we fix the ISRU productivity on the normal rate, the sensitivity of the IMLEO with respect to the spacecraft structure mass is shown in Fig. 18.

The spacecraft mass ratio shown in Fig. 18 is the spacecraft structure mass divided by the structure mass in baseline [i.e., Eq. (11)]. Because the spacecraft structure mass influences the propellant cost directly, with the increase of spacecraft structure mass, the total IMLEO increases significantly. When the spacecraft structure mass is low, there is a tradeoff between enlarging the spacecraft for ISRU deployment and using small spacecraft without ISRU. When the spacecraft is more than 20% heavier than normal, it is not worth building a large spacecraft for ISRU deployment. As a

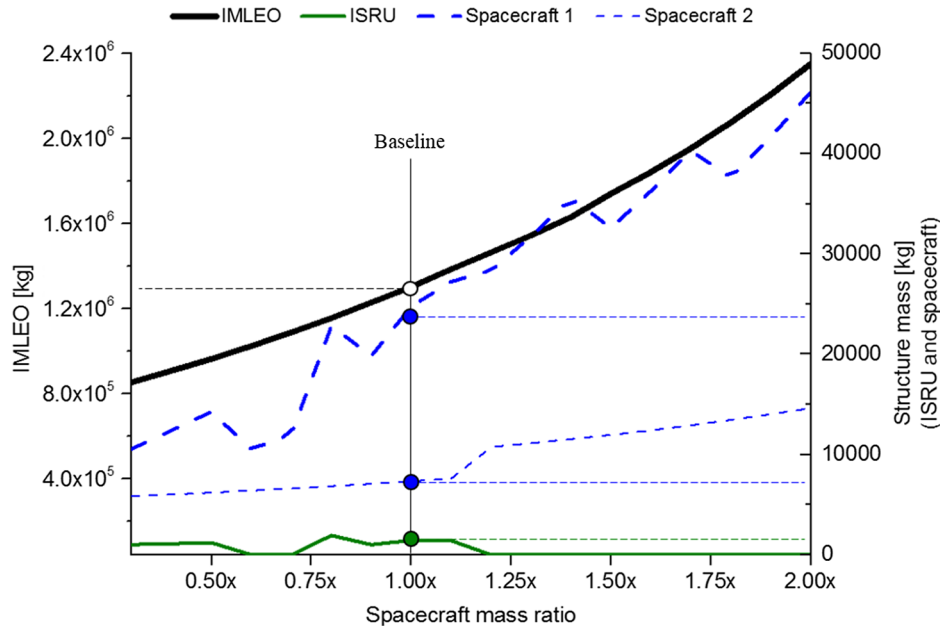


Fig. 18 Results of multi-mission lunar campaign design with respect to spacecraft structure mass (interval: one year).

result, the structure mass of small spacecraft starts to increase and the division of roles between two types of spacecraft becomes less clear. Both types of spacecraft start to deliver crews, equipment, and samples. Note that we can still observe that the structure mass of spacecraft 1 matches the variation pattern of the ISRU plant mass if ISRU is deployed in the mission, even though the structure mass is also increasing at the same time.

IV. Conclusions

This paper proposes two optimization methods to solve the space logistics problem considering nonlinear spacecraft and nonlinear ISRU models. First, the MILP optimization framework is developed, based on the GMCNF model. Piecewise functions are used to approximate the nonlinear spacecraft and ISRU models, as well as to convert the models into binary mixed-integer programming formulations. After linearizing the remaining quadratic terms, the whole problem is converted into a MILP problem. This MILP-based optimization framework can always find the global optimum of the problem with approximated nonlinear models. The second method is the SA-based optimization framework, which is developed with the concern that a higher-fidelity spacecraft model may be considered. In this method, the spacecraft model is separated from the mission planning part. This SA-based optimization framework can never guarantee or certify its global optimality.

To illustrate the effect of both optimization methods, a campaign-level mission including three lunar missions is optimized.

For the MILP method, the influences of mission interval are studied. The IMLEO of the space logistics system first decreases with the increase of mission interval until ISRU production cannot provide further benefits to the space logistics system. Moreover, even though the spacecraft are all designed as multiuse without specific functions identified initially, they are all assigned specific functions in the result. The large spacecraft are typically used to deliver equipment and ISRU plants, whereas the small ones are used to deliver crews and samples. Thus, the size of the large spacecraft is directly influenced by the mass of the ISRU plant deployed in the mission (see Figs. 15, 17, and 18).

The influences of ISRU and spacecraft model fidelities are also studied, as shown in Table 10. A linear ISRU or spacecraft model would not improve the computational efficiency because the nonlinear models can be linearized into binary mixed-integer

programming formulations and most of the variables and constraints come from the mission planning part, which is a GMCNF model. Therefore, the MILP method developed in this paper solves the space logistics problem, including nonlinear ISRU and spacecraft models without much deterioration of computational efficiency.

Besides the MILP method, a heuristic optimization algorithm based on SA is also proposed. Compared with the MILP method, the SA method cannot guarantee an optimal solution and the quality of the results is strongly dependent on the initial settings. For the problem considering medium-fidelity spacecraft model, the MILP method is always a better choice. However, the separated spacecraft model in the SA optimization framework makes it easier to consider a higher-fidelity spacecraft model, such as considering fuel types and specifically predefined functions of each type of spacecraft.

The ISRU productivity and spacecraft structure mass sensitivity are also studied. The IMLEO decreases with the increase of ISRU productivity. The spacecraft structure mass is also an important factor in determining the space logistics strategy. With the increase of spacecraft structure mass, IMLEO of the system increases significantly. When the spacecraft structure mass is too large, as compared with the normal condition, it is too expensive to build a large spacecraft for ISRU deployment and the functional difference between different types of spacecraft also become less clear.

Future work includes considering uncertainties in the transport scheme, which leads to flexible time windows. Moreover, the influences and interactions among different fuel types and ISRU plants deployed will make the problem much more complex. The piecewise linear approximation mentioned in this paper for linearizing nonlinear ISRU and spacecraft models is only suitable for a single variable function. To solve the space logistics problem considering higher-fidelity spacecraft and ISRU models by the MILP method, the formulation that converts the piecewise linear function into a binary mixed-integer programming formulation should be extended to a multivariable case.

Appendix: Parametric Spacecraft Model

This Appendix lists preexisting spacecraft parameters upon which the spacecraft model of Taylor were based [15]. Table A1 lists the spacecraft parameters, and Table A2 lists the fuel type and corresponding function values.

Table A1 List of spacecraft data used for spacecraft model [15]

Spacecraft name	Fuel identification (ID)	Propellant mass, kg	Payload mass, kg	Structural mass, kg
Saturn V third stage	2	107,725	0	12,014
Apollo command module (CM)	3	0	524	4,841
Apollo service module (SM)	3	18,413	60	6,053
Apollo LM DS	3	8,804	500	2,770
Apollo LM AS	3	2,358	250	1,719
Lunar crew exploration vehicle (CEV) CM	6	363	500	8,034
Lunar CEV SM	5	7,222	0	3,027
Altair DS	2	28,932	2,200	6,182
Altair AS	4	5,257	100	4,964
Earth departure stage (EDS)	2	226,693	0	22,500
Altair cargo carrier	0	0	15,000	1,000
Soyuz TM	3	900	255	7,250
Soyuz TMA	3	900	355	7,220
Progress M	3	900	2,350	7,450
Progress M1	3	900	1,800	7,150
Zvezda service module	0	0	10,000	20,000
STS-stage 2 (orbiter)	2	12,412	18,000	78,498
Soyuz-stage 2 (upper)	1	22,845	0	2,355
Proton-stage 2	3	46,562	0	4,115
ISS CEV CM (3 crew + cargo)	6	2,000	400	8,008
ISS CEV CM (6 crew)	6	2,000	0	8,079
ISS CEV (pressurized cargo)	0	0	3,500	7,683
ISS CEV SM	4	2,033	0	3,997
ATV: Automated transfer vehicle	5	2,613	5,500	10,470
HTV: H-II transfer vehicle	5	2,000	6,000	10,000
ISS CEV CM prop	5	2,000	400	8,008
EDS (75 mt)	2	129,500	0	19,986

Table A2 List of fuel and corresponding parameters used for spacecraft model [15]

Fuel type	ID	I_{sp} , s	α
LOX/kerosene	1	330	0.045
LOX/LH2	2	420	0.079
N2O4/UDMH	3	310	0.08
LCH4/LOX	4	318	0.958
MMH/N2O4	5	307	0.226
GOX/ethanol	6	300	3.9353

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